Rapport TP3 - Orchestration avec Docker Compose

Cloud Computing et Virtualisation - M1 SSII

Etablissement: Institut Superieur d'Informatique

Module: Cloud Computing et Virtualisation

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Objectifs

- Manipuler Docker Compose.
- Orchestrer deux services: Application Web et Base de donnees.

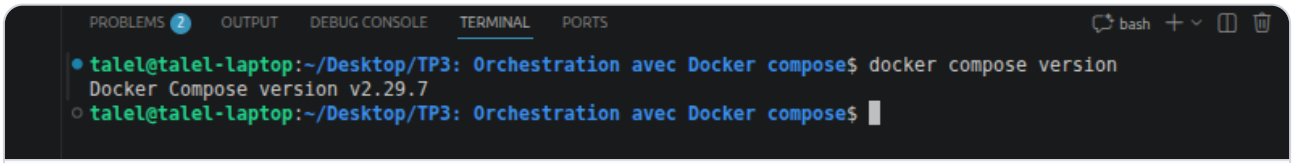
Partie 1 - Installation de Docker Compose

Commandes utilisees

```
sudo apt-get update
sudo apt-get install docker-compose-v2
docker compose version
```

Alternative manuelle (si le paquet est indisponible)

```
mkdir -p ~/.docker/cli-plugins
curl -SL https://github.com/docker/compose/releases/download/v2.29.7/docker-compose-linux-x86_64 -o ~/.docker/cli-plugins/docker-compose
chmod +x ~/.docker/cli-plugins/docker-compose
docker compose version
```



Legende: Verification de l'installation Docker Compose avec la commande docker compose version.

Partie 2 - Orchestration avec Docker Compose

Creation et demarrage des conteneurs

Arborescence attendue:

- sources/app/db-config.php
- sources/app/index.php
- sources/app/validation.php
- sources/db/articles.sql
- sources/db/Dockerfile
- docker-compose.yml

```
PROBLEMS 2   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   bash

talel@talel-laptop:~/Desktop/TP3: Orchestration avec Docker compose$ tree
├── docker-compose.yml
├── image.png
├── RAPPORT_TP3_Docker_Compose.md
├── README.md
├── sources
│   ├── app
│   │   ├── db-config.php
│   │   ├── Dockerfile
│   │   ├── index.php
│   │   └── validation.php
│   └── db
│       ├── articles.sql
│       └── Dockerfile
└── Tp3_OrchestrationWithDockerCompose.pdf

5 directories, 11 files
talel@talel-laptop:~/Desktop/TP3: Orchestration avec Docker compose$
```

Legende: Arborescence du projet (sources/app, sources/db, Dockerfile et docker-compose.yml).

```yaml
services:
  app:
    build:
      context: ./sources/app
    container_name: tp3_app
    ports:
      - "8080:80"
    depends_on:
      db:
        condition: service_healthy
    networks:
      - app_net

  db:
    build:
      context: ./sources/db
    container_name: tp3_db
    restart: unless-stopped
    environment:
      MYSQL_ROOT_PASSWORD: root
      MYSQL_DATABASE: test
      MYSQL_USER: test
      MYSQL_PASSWORD: test
    ports:
      - "3306:3306"
    healthcheck:
      test: ["CMD", "mysqladmin", "ping", "-h", "localhost", "-uroot", "-proot"]
      interval: 5s
      timeout: 5s
      retries: 30
      start_period: 30s
    volumes:
      - db_data:/var/lib/mysql
    networks:
      - app_net
```

Legende: Contenu du fichier docker-compose.yml (services app et db, reseau et volume).

```
urces > app > db-config.php > ...
1  <?php
2  const DB_DSN = 'mysql:host=db;dbname=test';
3  const DB_USER = "test";
4  const DB_PASS = "test";
5  $options = array(
6      PDO::MYSQL_ATTR_INIT_COMMAND => "SET NAMES utf8", // encodage utf-8
7      PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION, // gérer les erreurs en tant qu'exception
8      PDO::ATTR_EMULATE_PREPARES => false // faire des vrais requêtes préparées et non une émulation
9  );
```

Legende: Fichier db-config.php avec host=db pour la connexion inter-conteneurs.

```
talel@talel-laptop:~/Desktop/TP3: Orchestration avec Docker compose$ docker compose up -d --build
 => => transferring dockerfile: 112B                                                                    0.0s
 => [db internal] load metadata for docker.io/library/mysql:5.7                                         2.2s
 => [db internal] load .dockerignore                                                                    0.2s
 => => transferring context: 2B                                                                         0.0s
 => [db internal] load build context                                                                    0.2s
 => => transferring context: 34B                                                                        0.0s
 => [db 1/2] FROM docker.io/library/mysql:5.7@sha256:4bc6bc963e6d8443453676cae56536f4b8156d78bae03c0145cbe47c2aad73bb  0.0s
 => CACHED [db 2/2] COPY articles.sql /docker-entrypoint-initdb.d/articles.sql                          0.0s
 => [db] exporting to image                                                                             0.4s
 => => exporting layers                                                                                 0.0s
 => => writing image sha256:09921aaa1b0342cca0d4ba016702db6722d4cf524fba5e2e6d7fe7bff0c8d139            0.1s
 => => naming to docker.io/library/tp3orchestrationavecdockercompose-db                                 0.1s
 => [db] resolving provenance for metadata file                                                         0.0s
 => [app internal] load build definition from Dockerfile                                                0.3s
 => => transferring dockerfile: 143B                                                                    0.0s
 => [app internal] load metadata for docker.io/library/php:8.2-apache                                   1.9s
 => [app internal] load .dockerignore                                                                   0.6s
 => => transferring context: 2B                                                                         0.0s
 => [app 1/4] FROM docker.io/library/php:8.2-apache@sha256:2d3b5ac95865397e61a6129fb2c70f676a79eafd855e5ff74ff0b409a33dcfe5  0.0s
 => [app internal] load build context                                                                   0.3s
 => => transferring context: 127B                                                                       0.0s
 => CACHED [app 2/4] RUN docker-php-ext-install pdo pdo_mysql                                           0.0s
 => CACHED [app 3/4] WORKDIR /var/www/html                                                              0.0s
 => CACHED [app 4/4] COPY . /var/www/html                                                               0.0s
 => [app] exporting to image                                                                            0.3s
 => => exporting layers                                                                                 0.0s
 => => writing image sha256:6608056c61332bca027325512f8924efc925c0b1d75112e176e8b48aebeeb0b6            0.1s
 => => naming to docker.io/library/tp3orchestrationavecdockercompose-app                                0.1s
 => [app] resolving provenance for metadata file                                                        0.0s
[+] Running 2/2
 ✔ Container tp3_db   Healthy                                                                           0.5s
 ✔ Container tp3_app  Running                                                                           0.0s
talel@talel-laptop:~/Desktop/TP3: Orchestration avec Docker compose$
```

Legende: Execution de docker compose up -d --build pour creer et demarrer les conteneurs.

```
97
98   Resultat:
99   - La page Articles s'affiche correctement.
100
101  Capture a inserer:
```

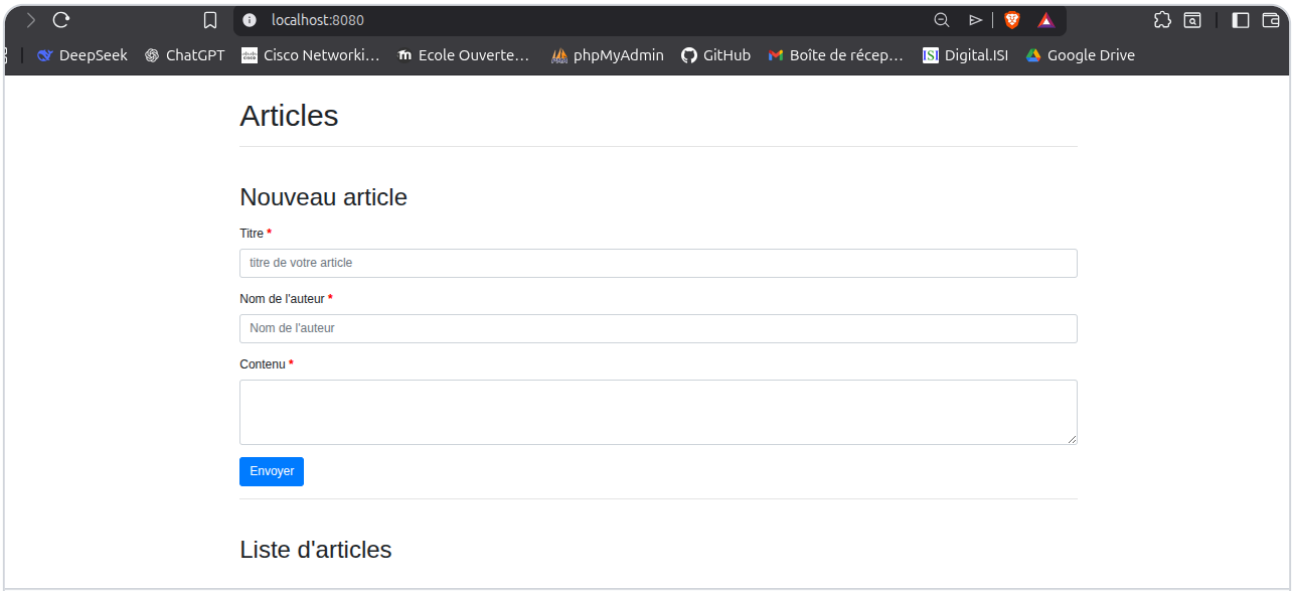
```
PROBLEMS 2   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                                  bash

talel@talel-laptop:~/Desktop/TP3: Orchestration avec Docker compose$ docker compose ps
NAME      IMAGE                                COMMAND                  SERVICE   CREATED         STATUS                   PORTS
tp3_app   tp3orchestrationavecdockercompose-app   "docker-php-entrypoi…"   app       24 minutes ago   Up 24 minutes            0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
tp3_db    tp3orchestrationavecdockercompose-db    "docker-entrypoint.s…"   db        24 minutes ago   Up 24 minutes (healthy)  0.0.0.0:3306->3306/tcp, :::3306->3306/tcp, 330
60/tcp
talel@talel-laptop:~/Desktop/TP3: Orchestration avec Docker compose$
```

Legende: Etat des services via docker compose ps (app Up et db Up/healthy).

Test de la page web

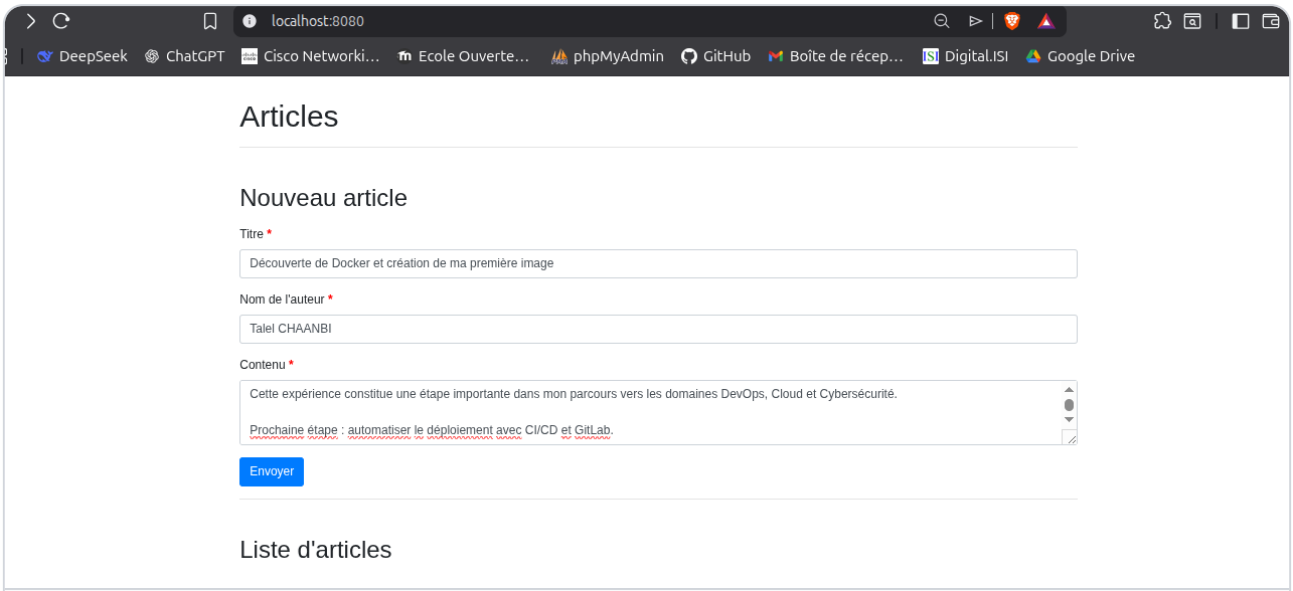
URL testee: http://localhost:8080

Resultat: la page Articles s'affiche correctement.

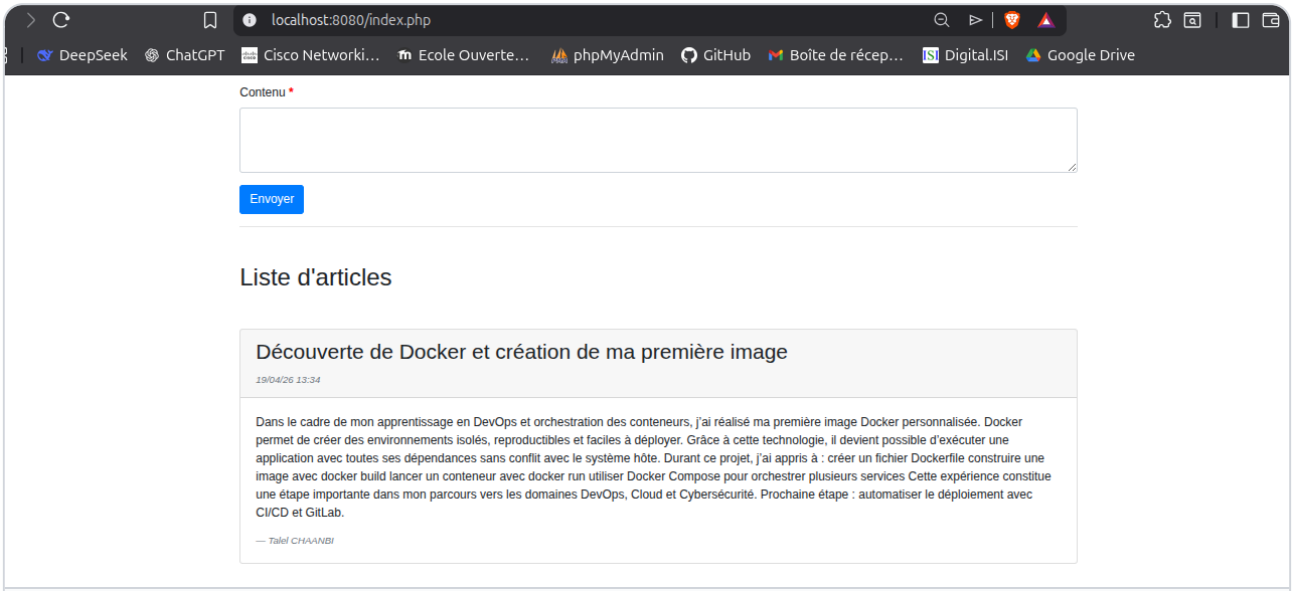


Legende: Affichage de la page web Articles sur http://localhost:8080.

Ajout d'un article



Legende: Formulaire de nouvel article rempli (titre, auteur, contenu).

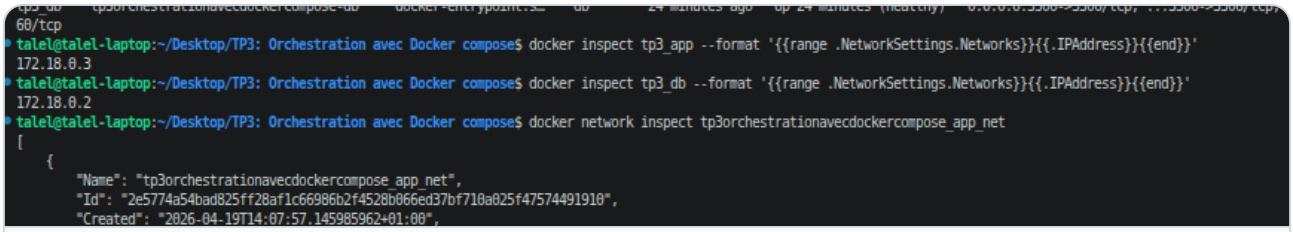


Legende: Verification de l'ajout de l'article dans la liste.

Adresses IP et reseau app_net

```
docker inspect tp3_app --format '{{range .NetworkSettings.Networks}}
{{.IPAddress}}{{end}}'
docker inspect tp3_db --format '{{range .NetworkSettings.Networks}}
{{.IPAddress}}{{end}}'
docker network inspect tp3orchestrationavecdockercompose_app_net
```

- IP conteneur app: 172.18.0.3
- IP conteneur db: 172.18.0.2
- Reseau app_net: bridge, 172.18.0.0/16, gateway 172.18.0.1



Legende: Adresses IP des conteneurs obtenues avec docker inspect.

```
talel@talel-laptop:~/Desktop/TP3: Orchestration avec Docker compose$ docker network inspect tp3orchestrationavecdockercompose_app_net
[
    {
        "Name": "tp3orchestrationavecdockercompose_app_net",
        "Id": "2e5774a54bad825ff28af1c66986b2f4528b066ed37bf710a025f47574491910",
        "Created": "2026-04-19T14:07:57.145985962+01:00",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.18.0.0/16",
                    "Gateway": "172.18.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {
            "46d0007f0bc674d2cd154a8ee6014f7342fa2c61651032b3a40958043dbac8db": {
                "Name": "tp3_db",
                "EndpointID": "3f0c2b8cd44b18c2bbcb085088c7d467204a880c530f865f44b33680df2e7d9e",
                "MacAddress": "fa:57:59:85:4d:38",
                "IPv4Address": "172.18.0.2/16",
                "IPv6Address": ""
            },
            "e50801cb6e4059c099814cf699c18177ffbf071e8fb6801c4bf2b92725f48090": {
                "Name": "tp3_app",
                "EndpointID": "3ec91824c1d68274a3107250ca3517f78b154cf5425e6180cfdc66c1dcc47977",
```

Legende: Details du reseau app_net obtenus avec docker network inspect.

Conclusion

Ce TP a permis de mettre en place une orchestration multi-conteneurs avec Docker Compose, de separer l'application web et la base de donnees, de valider la communication via app_net, et de verifier l'ajout d'articles avec persistance des donnees.